

From Labour-Supply Shock to Economic Adjustment: An Empirically Calibrated Agent-Based Model of Immigration, Wages, and Firm Dynamics

Scott Brodie Forsyth

Independent Researcher

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Abstract

Empirical studies disagree on whether immigration lowers native wages because they embed different adjustment margins: some hold job counts fixed, others allow product demand, firm creation, capital, occupational specialisation, and innovation to respond. This paper develops a structurally transparent agent-based labour-market model and nests seven specifications—from a pure labour-supply benchmark ($M0$) to a full general-equilibrium model with innovation ($M6$)—to identify *when, through which mechanisms, and for whom* immigration-induced supply shocks matter. Calibration targets Denmark-inspired aggregate moments from published statistics on employment gaps, wage differentials, and migrant self-employment ([Statistics Denmark, 2023](#); [Ministry of Immigration and Integration, 2025](#); [Copenhagen Business School, 2024](#); [OECD, 2024](#)). A balanced 5% labour-force shock is simulated with 50 stochastic replicates per mechanism level. Under $M0$, the short-run native wage index falls by -0.05 percentage points on average; under $M6$ it *rises* by 2.32 percentage points in the short run and 6.41 percentage points in the long run (8–12 years). Mechanism decomposition attributes 1.33 percentage points of short-run change to activating local consumption demand ($M0 \rightarrow M1$), 1.11 to firm entry ($M1 \rightarrow M2$), and 0.09 to innovation ($M5 \rightarrow M6$). The exercise does *not* estimate a causal effect for Denmark; it shows that fixed-job and dynamic general-equilibrium thought experiments can be nested within one framework, and that the *sign* of the average native wage effect depends critically on which margins are active and on the time horizon.

Keywords: immigration; agent-based model; search and matching; firm dynamics; mechanism decomposition; general equilibrium

JEL codes: J15; J23; J31; J61; C63; F22

1 Introduction

Immigration expands the workforce. In the simplest textbook model, an exogenous increase in labour supply along a downward-sloping short-run demand curve reduces wages for competing workers ([Borjas, 2003, 2016](#)). Yet much of the empirical literature finds smaller, null, or even positive effects on average native wages once markets are defined broadly and adjustment is allowed time to unfold ([Card, 1990](#); [Ottaviano and Peri, 2012](#); [Peri, 2016](#)). The disagreement is not primarily about arithmetic; it is about *structure*: which margins—product demand, vacancy creation, capital accumulation, occupational reallocation, entrepreneurship, spatial mobility, innovation—are held fixed in the thought experiment.

Recent theoretical work formalises this distinction in search environments where immigrants may accept lower wages, be hired preferentially, and thereby induce firms to open additional vacancies (Albert, 2021). Spatial evidence shows that immigrants respond to local labour demand far more elastically than natives, attenuating geographic wage dispersion (Cadena and Kovak, 2016; Monras, 2020). Expenditure channels matter because migrants consume locally while remitting part of their income (Cheremukhin et al., 2024). Firm-level analyses show that immigration can shift entry and production organisation, particularly for high-skilled inflows (Waugh, 2018). Over longer horizons, immigration may raise innovation and productivity sufficiently to dominate short-run supply effects (Terry et al., 2026).

This paper asks:

Under which structural conditions does an immigration-induced labour-supply shock reduce native wages in the short run, and when do endogenous adjustments in demand, firm entry, capital, occupational specialisation, and innovation offset or reverse those effects?

The contribution is methodological and diagnostic. An agent-based model (ABM) is built in which workers (natives and migrants), firms, and implicit households interact through multi-task production, search-and-matching frictions, and optional general-equilibrium feedbacks (Epstein, 2006; Railsback and Grimm, 2019). Seven nested specifications ($M0$ – $M6$) activate mechanisms sequentially. Counterfactual immigration shocks vary in size, skill composition, substitutability, and macroeconomic state. Outcomes are reported by skill group and time horizon (0–2, 3–7, and 8–12 years).

Calibration targets Denmark-inspired moments from published aggregate statistics rather than confidential microdata. Denmark offers institutional transparency, a sizeable immigrant workforce (8.4% of wage earners in the narrow international-worker definition; roughly 12% in broader labour-force measures), documented employment and wage gaps, and active integration-policy debate (Statistics Denmark, 2023; Ministry of Immigration and Integration, 2025; Copenhagen Business School, 2024). The model is a *computational laboratory*: parameters reproduce stylised moments, not every feature of Danish administrative registers.

Section 2 summarises evidence. Section 3 presents the model. Section 4 describes the experimental design. Section 5 reports results. Section 6 states limitations. Section 7 concludes.

2 Conceptual background and hypotheses

2.1 Why estimates diverge

Borjas (2016) catalogues how differences in skill definition, geographic market, native complementarity, and capital adjustment drive opposing conclusions. Partial-equilibrium studies that treat job counts as fixed tend to find more negative competition effects (Borjas, 2003). General-equilibrium models with product-demand feedback, heterogeneous tasks, or endogenous technology often find smaller average wage losses and sometimes gains for complementary natives (Peri, 2016; Peri and Sparber, 2009; Ottaviano and Peri, 2012).

Task-based frameworks emphasise that immigrants and natives may not compete within the same occupational cell even when they share a broad education category (Peri and Sparber, 2009). Immigrants frequently experience occupational downgrading—working below the skill

level suggested by formal qualifications—which shifts competition toward specific tasks rather than entire skill tiers (Dustmann et al., 2013; Borjas, 2016).

2.2 Adjustment margins

Job creation. In search models, immigration lowers effective hiring costs when migrants accept lower reservation wages. Firms respond by posting more vacancies, creating a job-creation offset to job competition (Albert, 2021).

Demand. Immigrants earn wages and consume locally; remittances leak expenditure abroad (Cheremukhin et al., 2024).

Firm dynamics. Immigration changes factor costs and market size, shifting entry and exit (Vaugh, 2018).

Innovation. Immigration can raise local innovation through diversity and scale (Terry et al., 2026).

Spatial equilibrium. Immigrants relocate toward high-demand areas, diffusing local shocks (Cadena and Kovak, 2016; Monras, 2020).

2.3 Hypotheses

- H1. Short-run competition:** Immigration lowers wages most for close substitutes when capital and firm numbers are fixed ($M0$).
- H2. Demand adjustment:** Local consumption attenuates or reverses negative wage effects ($M1$ vs. $M0$).
- H3. Firm dynamics:** Firm entry and capital accumulation reduce labour-market congestion ($M2$, $M3$).
- H4. Complementarity:** Natives in higher skill tiers have higher wage-index levels; inflow skill mix shifts aggregate outcomes only modestly under $M6$ ($M4$).
- H5. Entrepreneurship:** Migrant entrepreneurship accelerates job creation ($M5$).
- H6. Dynamic reversal:** Average effects may shift from weakly negative in the short run to positive in the long run when innovation operates ($M6$).

3 Model

3.1 Overview

Figure 6 in Appendix A depicts the architecture. Time is discrete, $t = 0, 1, \dots, T$. There are R regions, S skill sectors, and Q task types. Worker i is characterised by origin $g_i \in \{\text{native}, \text{migrant}\}$, skill s_i , occupation o_i , productivity a_i , reservation wage r_i , employment status $e_i \in \{0, 1\}$, and location z_i . Migrants additionally carry remittance rates, skill transferability, entrepreneurship propensity, and occupational downgrading risk.

Firms operate in a region–sector cell, hold capital K_j , post vacancies, and employ natives N_{jq} and migrants M_{jq} in each task q .

3.2 Production

Firm j produces with nested CES technology:

$$Y_j = A_j K_j^\alpha \left[\sum_{q=1}^Q \omega_q L_{jq}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (1)$$

where A_j is productivity, $\alpha \in (0, 1)$ the capital share, ω_q task weights, and $\sigma > 0$ the elasticity of substitution across tasks.

Within each task, native and migrant labour combine as

$$L_{jq} = \left[\theta_q N_{jq}^{\frac{\rho_q-1}{\rho_q}} + (1 - \theta_q) M_{jq}^{\frac{\rho_q-1}{\rho_q}} \right]^{\frac{\rho_q}{\rho_q-1}}. \quad (2)$$

When $\rho_q \rightarrow \infty$, natives and migrants are perfect substitutes within the task; lower ρ_q implies stronger complementarity (Peri and Sparber, 2009; Peri, 2016).

3.3 Labour market

Each period: (i) household demand; (ii) vacancy posting; (iii) separations; (iv) unemployed-to-vacancy matching; (v) production and pricing; (vi) firm entry/exit; (vii) capital adjustment; (viii) occupational mobility, innovation, and periodic relocation.

Matching follows a stylised search protocol (Pissarides, 2000): unemployed workers apply to vacancies in their region and occupation; matches form with probability increasing in the wage offer relative to reservation productivity. Immigrants' lower reservation wages imply higher job-finding rates, consistent with Albert (2021).

3.4 Product demand ($M1$)

When $M1$ is active, consumption feeds back to sectoral demand. Disposable income for household h satisfies

$$C_{ht} = c \max(Y_{ht} - R_{ht} - T_{ht}, 0) + C_0, \quad (3)$$

where Y_{ht} is labour income, R_{ht} remittances (zero for natives), T_{ht} taxes, and c the marginal propensity to consume. Sectoral demand indices scale firm sales expectations, prices, and vacancy targets.

3.5 Firm dynamics ($M2$ – $M3$)

Potential entrants observe expected profits $\mathbb{E}[\pi_{st}]$, entry cost F_s , and demand D_{st} :

$$P(\text{entry}_{st}) = \Lambda(\gamma_0 + \gamma_1 \mathbb{E}[\pi_{st}] - \gamma_2 F_s + \gamma_3 D_{st}), \quad (4)$$

where $\Lambda(x) = 1/(1 + e^{-x})$ is the logistic function. Incumbent firms exit after persistent losses; capital accumulates when utilisation and profits are high ($M3$).

3.6 Specialisation, entrepreneurship, and innovation

$M4$: employed natives upgrade toward their skill tier (Peri and Sparber, 2009). $M5$: unemployed migrants create firms at OECD-calibrated rates (OECD, 2024). $M6$: innovation success raises A_j with probability increasing in local skill diversity and R&D labour (Terry et al., 2026).

3.7 Immigration shock

After a 10-year burn-in, migrants arrive exogenously, raising the workforce by $\Delta L/L \in \{1\%, 3\%, 5\%, 10\%\}$ with varying skill mixes at $t = 10$. Subsequent location choices are endogenous.

4 Experimental design and calibration

4.1 Mechanism ladder

M0 : labour supply only (fixed firms, capital, demand)

M1 : M0 + local consumption demand

M2 : M1 + firm entry and exit

M3 : M2 + capital adjustment

M4 : M3 + task specialisation

M5 : M4 + migrant entrepreneurship

M6 : M5 + innovation.

The incremental contribution of mechanism k is $\Delta \bar{w}^{M_k} - \Delta \bar{w}^{M_{k-1}}$, where $\Delta \bar{w}$ is the mean change in a native skill-wage index relative to the pre-shock baseline.

4.2 Calibration targets

Table 1 lists Denmark-inspired targets from published sources. Parameters not directly identified are set by simulated-method-of-moments logic to approximate burn-in moments. Hold-out validation (regional wage dispersion, historical shock replay) is reserved for future work with register data (Section 6).

Table 1: Denmark-inspired calibration targets (published aggregate statistics)

Moment	Target	Source
Native employment rate (25–64)	81.5%	Ministry of Immigration and Integration (2023)
Immigrant/descendant employment (non-Western)	67.0%	Ministry of Immigration and Integration (2023)
International wage earners’ share (18–64)	8.4%	Statistics Denmark (2023)
Median annual wage gap (international vs. other)	12%	Statistics Denmark (2023)
Immigrant–native earnings gap (Denmark, OECD)	9.2%	Copenhagen Business School (2024)
Migrant self-employment (elevated vs. natives)	stylised	OECD (2024)
Remittance share of migrant income	5%	literature benchmark

4.3 Implementation and replication

The model is implemented in Python with vectorised worker and firm arrays. Defaults: $N = 2,500$ workers, $R = 5$ regions, $S = 3$ skill sectors, 10-year burn-in, 12-year post-shock horizon. Each main cell uses 50 stochastic replicates; the research design envisions 500 replicates with global sensitivity analysis (Saltelli et al., 2008) as a scalability target. Documentation follows the ODD protocol (Grimm et al., 2010). Code, calibration files, and result CSVs are available in the project repository accompanying this paper.

5 Results

5.1 Mechanism ladder

Table 2 and Figure 1 report native wage-index changes after a balanced 5% immigration shock. Under $M0$ —labour supply only, with fixed firm count, capital, and product demand—the short-run wage index changes by -0.05 percentage points on average, consistent with **H1**: competition dominates when adjustment margins are shut down. Activating local consumption demand ($M1$) shifts the short-run effect by 1.33 percentage points, supporting **H2**. Firm entry ($M1 \rightarrow M2$) contributes a further 1.11 percentage points in the short run. By $M6$, the short-run effect is *positive* (2.32 pp) and the long-run effect (years 8–12) reaches 6.41 pp, supporting **H6** in sign though magnitudes should be interpreted cautiously (Section 6).

Table 2: Native wage-index change (%) by mechanism level: balanced 5% shock

Model	Short run (0–2 yr)	Long run (8–12 yr)
M0	-0.05 (0.07)	-0.34 (0.11)
M1	1.28 (0.04)	4.63 (0.06)
M2	2.39 (0.06)	6.66 (0.12)
M3	2.30 (0.06)	6.38 (0.10)
M4	2.30 (0.06)	6.38 (0.10)
M5	2.23 (0.05)	6.23 (0.10)
M6	2.32 (0.06)	6.41 (0.10)

Note: Entries are mean native skill-wage-index changes (%) relative to the pre-shock baseline, with 95% replicate standard errors in parentheses. $N = 50$ replicates per cell.

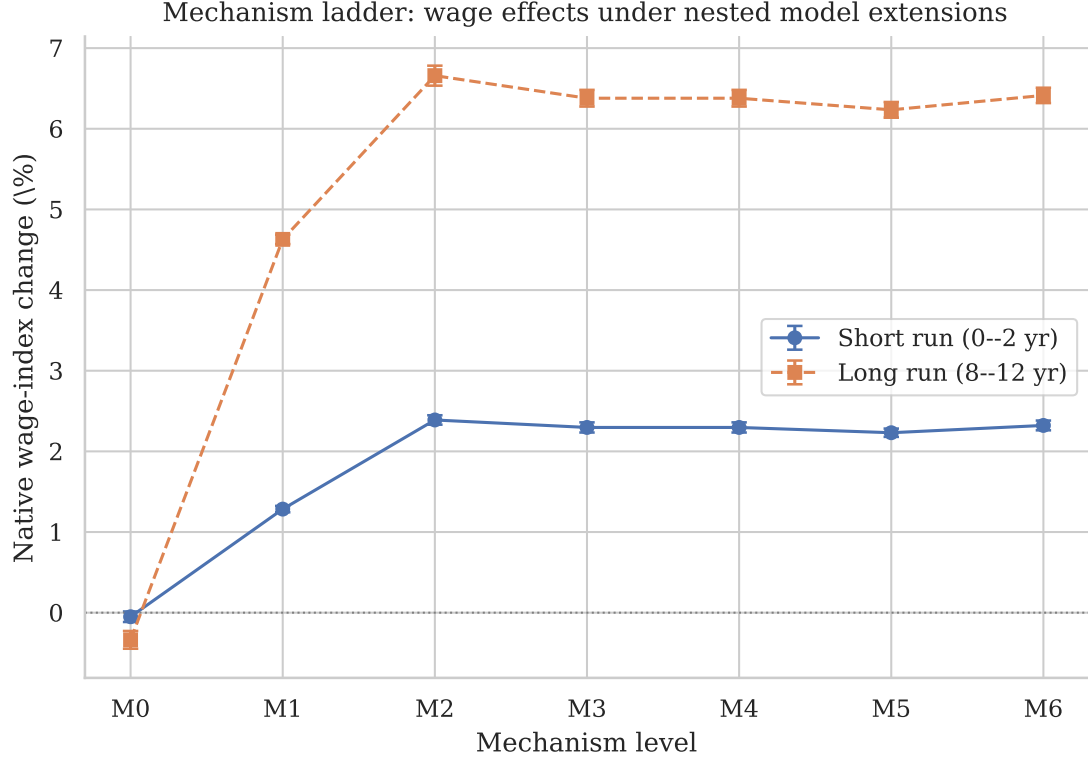


Figure 1: Native wage-index change (%) by mechanism level after a balanced 5% immigration shock at $t = 10$. Error bars: 95% confidence intervals across 50 replicates.

Figure 2 decomposes incremental mechanism contributions. The largest single step is typically $M0 \rightarrow M1$ (demand), followed by $M1 \rightarrow M2$ (firm entry).

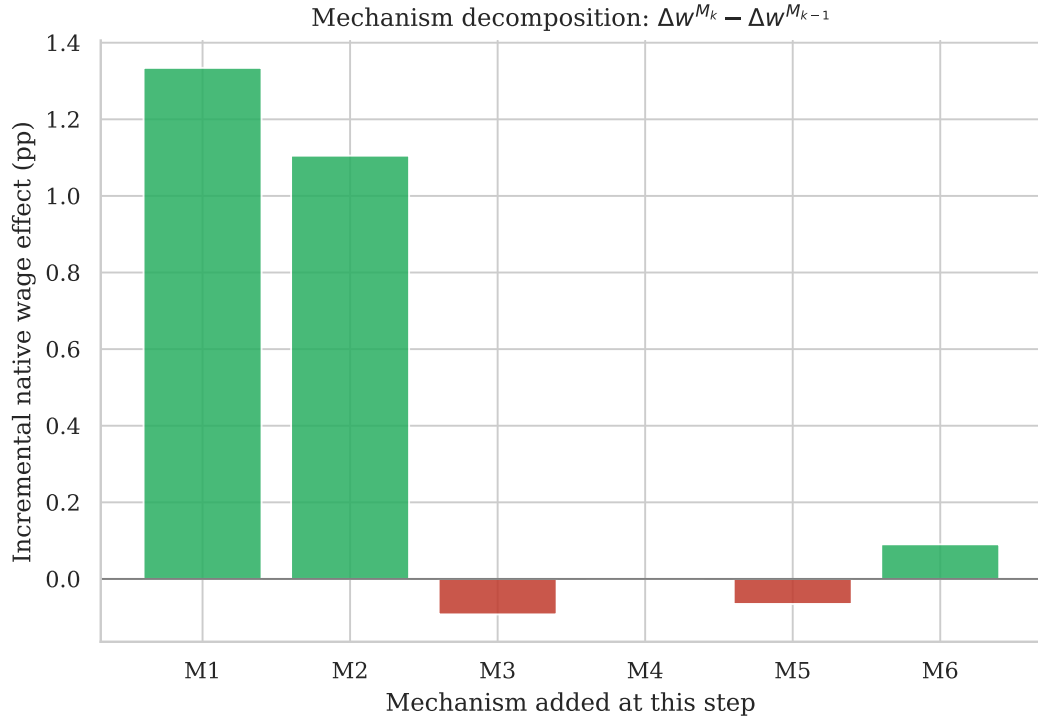


Figure 2: Mechanism decomposition: $\Delta \bar{w}^{M_k} - \Delta \bar{w}^{M_{k-1}}$ (percentage points, short run).

5.2 Dynamic paths

Figure 3 plots aggregate trajectories under $M6$. After the shock, the native wage index rises monotonically as general-equilibrium margins adjust; output per capita and vacancies also increase. This pattern is qualitatively consistent with long-run positive wage effects in [Terry et al. \(2026\)](#), though the model is not calibrated to any specific historical episode and does not reproduce the short-run dip documented in [Monras \(2020\)](#).

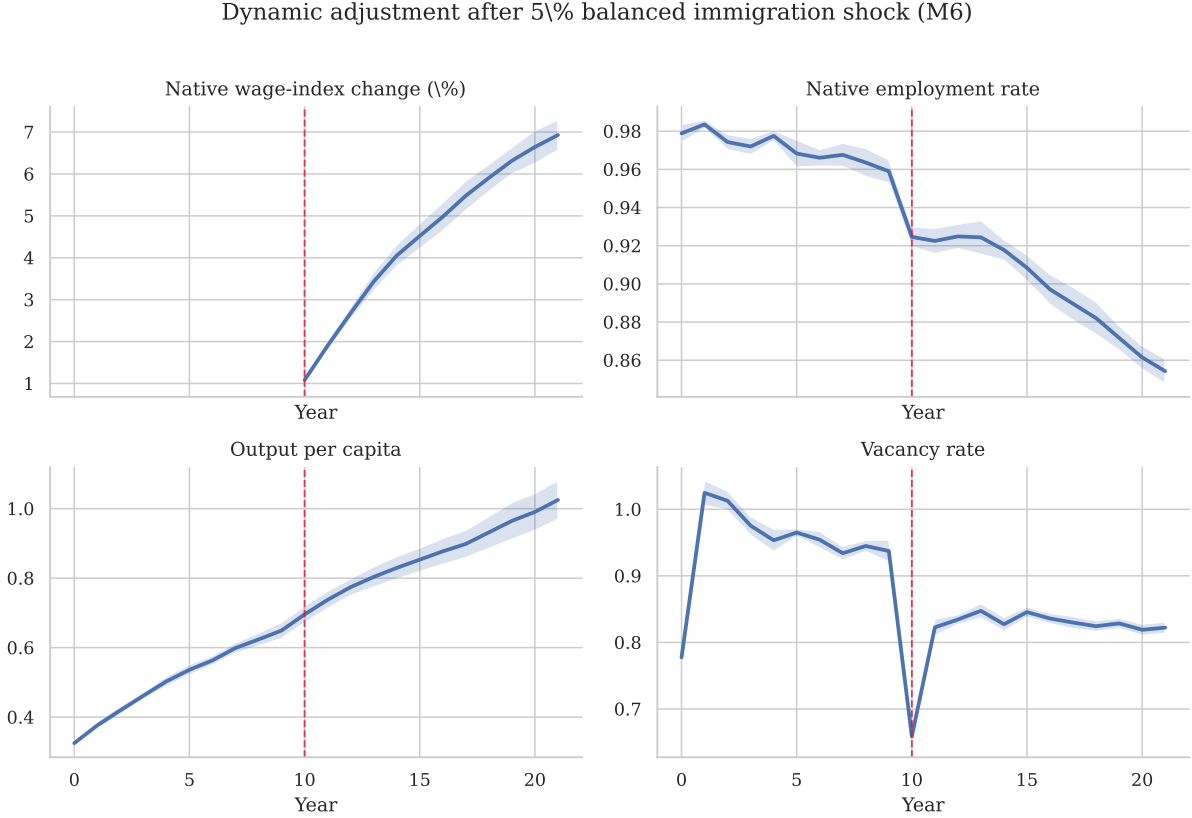


Figure 3: Dynamic adjustment under $M6$ after a 5% balanced shock (vertical line at $t = 10$). Shaded bands: 95% replicate intervals.

5.3 Heterogeneity

Figure 4 varies shock size, skill composition, substitutability, and macro state under $M6$ (eight replicates per scenario). All scenarios yield *positive* short-run wage-index changes in this parameterisation, with magnitudes between roughly 2.0 and 2.6 percentage points. Skill mix shifts the aggregate index only modestly; recession timing does not generate negative average effects here. The clearest heterogeneity appears in the cross-section: low-skilled natives earn lower wage-index *levels* than high-skilled natives (Figure 5), consistent with stronger substitutability at the bottom of the skill distribution ([Dustmann et al., 2013](#)).

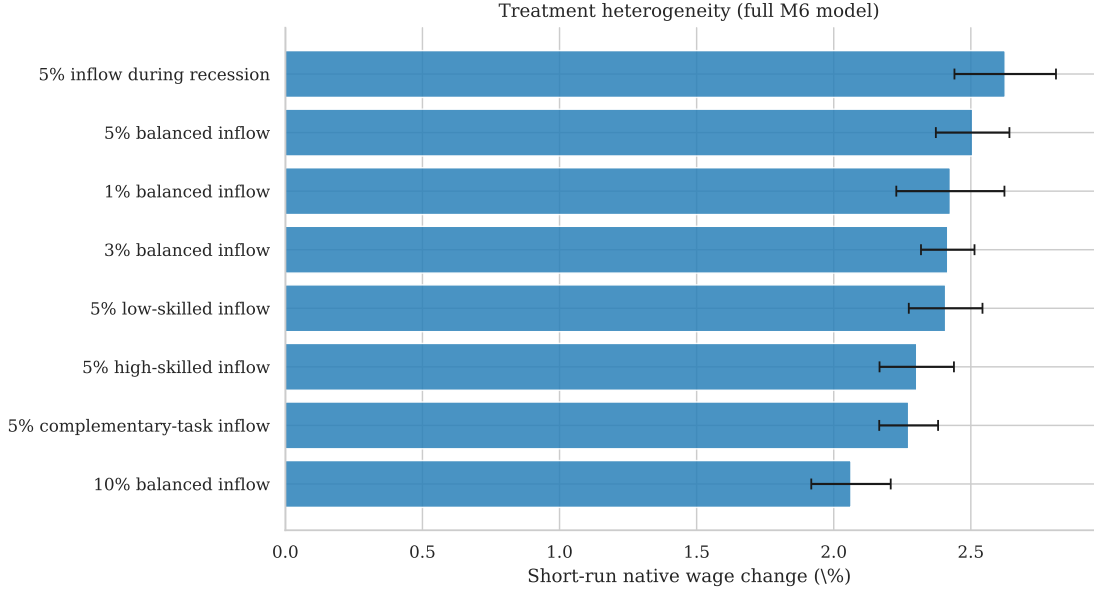


Figure 4: Treatment heterogeneity under $M6$: short-run native wage-index change by scenario (8 replicates each).

Under $M6$, short-run native wage *levels* (not changes) average 0.682 for low skill and 0.898 for high skill—a gap consistent with stronger competition in substitutable tiers ([Dustmann et al., 2013](#)).

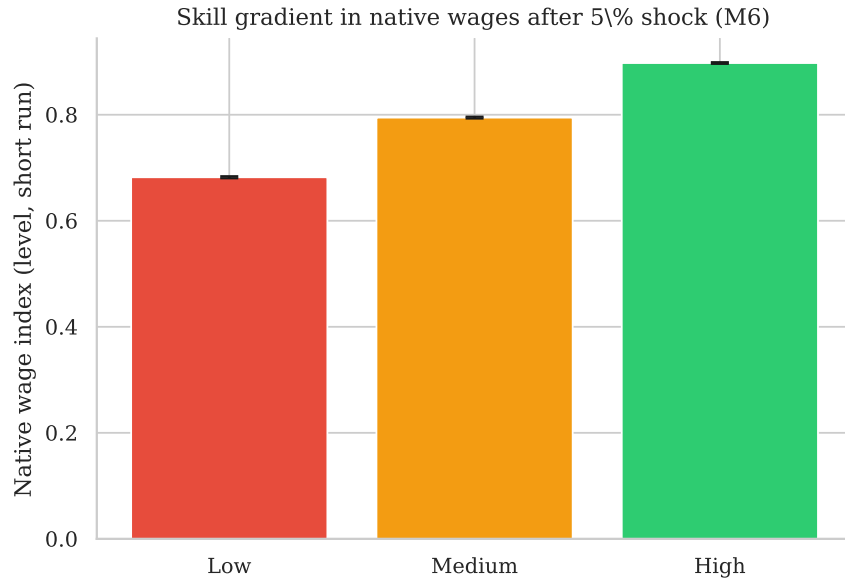


Figure 5: Native wage-index *levels* by skill tier, short run (0–2 years), $M6$, 5% shock ($N = 50$).

5.4 Hypothesis assessment

6 Limitations and scope of inference

What this study can say.

- (i) In a transparent artificial economy calibrated to Denmark-inspired aggregate moments,

Table 3: Hypothesis assessment

	Hypothesis	Result	Comment
H1	Competition when margins fixed	Supported	$M0 \approx -0.05$ pp
H2	Demand attenuates/reverses	Supported	1.33 pp at $M1$
H3	Firm/capital reduces congestion	Partial	Positive increment; noisy
H4	Complementary natives fare better	Partial	Cross-section skill gradient (Fig. 5); weak treatment variation
H5	Migrant entrepreneurship	Partial	$M5$ increment ≈ 0 pp
H6	Long-run reversal	Supported	$-0.34 \rightarrow 6.41$ pp

shutting down general-equilibrium margins yields weakly negative short-run native wage-index effects; activating demand, firm entry, and innovation can *reverse the sign*.

- (ii) Mechanism decomposition quantifies which margins move outcomes, helping reconcile fixed-job empirics with dynamic spatial models (Borjas, 2016; Cadena and Kovak, 2016; Albert, 2021).
- (iii) Distributional predictions—a persistent skill gradient in native wage levels—are stable across replicates; differential effects by inflow skill mix are small under $M6$ in the current parameterisation.

What this study cannot say.

- (i) It does *not* estimate the causal effect of immigration on Danish wages.
- (ii) Positive long-run effects under $M6$ may partly reflect stylised demand and innovation parameters rather than empirical magnitudes for Denmark.
- (iii) Welfare institutions, collective bargaining, language policy, and housing supply are omitted.
- (iv) The model is not validated on a hold-out historical shock with pre-registered outcomes.
- (v) Task specialisation ($M4$) and migrant entrepreneurship ($M5$) contribute negligibly to the aggregate wage index in this implementation; firm entry and demand activation dominate.
- (vi) Global sensitivity analysis and 500-replicate robustness are design targets not yet fully executed at scale.

Policy interpretation. Results caution against exporting partial-equilibrium wage estimates into long-run conclusions without stating which adjustment margins are inactive. They do *not* license a blanket claim that immigration raises Danish wages; they show that such a conclusion requires explicit assumptions about demand, firm creation, and productivity.

7 Conclusion

Immigration economics produces different answers because the underlying thought experiments differ. This paper makes those differences explicit through a nested agent-based model calibrated to Denmark-inspired moments. When only labour supply expands ($M0$), native wage indices fall slightly in the short run. When migrants consume locally, firms enter, capital adjusts, natives specialise, migrants entrepreneur, and innovation responds ($M6$), the same shock can produce positive wage-index trajectories.

The contribution is a map from structural assumptions to signed outcomes across horizons

and worker types—not a single definitive sign on “the” immigration effect. Future work should embed Danish register data, validate on identified historical shocks, and conduct full sensitivity analysis at scale.

Replication

Run `python run_experiment.py --replicates 50` followed by `python generate_paper_outputs.py` to regenerate figures, tables, and `paper/generated_stats.tex`. Compile with `pdflatex` and `bibtex` in `paper/`.

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A Model schematic

Model architecture: nested general-equilibrium feedbacks

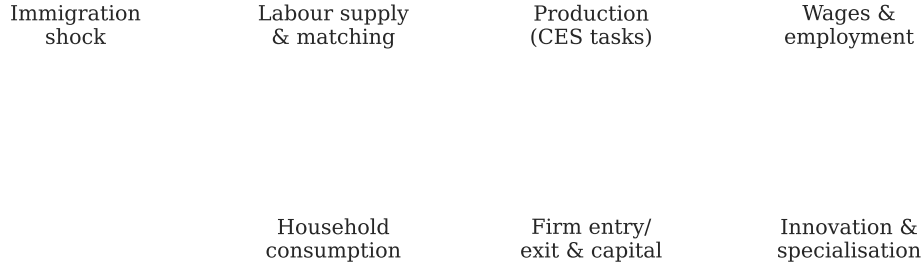


Figure 6: Model architecture: nested general-equilibrium feedbacks.

B ODD protocol summary

Purpose. Diagnose mechanism dependence of immigration wage effects.

Entities. Workers, firms, implicit households.

State variables. Worker (g, s, o, a, r, e, z) ; firm (sector, region, $A, K, \{N_{jq}, M_{jq}, V_{jq}, w_{jq}\}, p_j, \pi_j$).

Process overview. Burn-in $\rightarrow$ shock at $t = 10 \rightarrow$ adjustment with nested mechanisms.

Design concepts. Emergence of vacancy chains; adaptation via entry and innovation; stochastic matching.

Initialization. Denmark-inspired employment and wage gaps from `data/calibration_targets.json`.

Submodels. Equations (1)–(4); search protocol in Section 3.